

Set Theory

Some Basic definition of sets with examples

	Definition	Example
Set	A well-defined collection of distinct objects is called a set.	$A = \{1,2,3,4,5\}$ A = set of a first five Natural Numbers. $A = \{x \in \mathbb{N} \mid \mathbb{N} \leq \mathbb{N}\}$
Objects	The objects are called the elements or members of the set, and it represents by small letter.	$A = \{1,2,3,4,5\}$ In the above set A, 1,2,3,4, and 5 are the elements of set.
Set Name	Sets are denoted by capital letters A,B,C ... X,Y,Z.	$A = \{1,2,3,4,5\}$ In the above set A is the name of the set.

Set Representation

A set can be representing in three different way

Descriptive form	Tabular form	Set builder notation
Listing all the elements of a set, separated by commas and enclosed within braces or curly brackets {}.	Stating in words the elements of the set.	Writing in symbolical form the common elements shared by all elements
$A = \{1,2,3,4,5\}$ $B = \{2,4,6,8,\dots,50\}$	A = set of a first five Natural Numbers. B = set of positive even integers less or equal to fifty	$A = \{x \in \mathbb{N} \mid \mathbb{N} \leq \mathbb{N}\}$ $A = \{y \in \mathbb{E} \mid 0 < y \leq 50\}$
Subset	if every element of A is the element of B $A \subseteq B$	$A = \{1,3,5\}$ $B = \{1,2,3,4,5\}$ $A \subseteq B = \{1, 3, 5\}$
Super Set	When $A \subseteq B$ then B is the superset of A.	
Proper Subset	If every element of A is the part of B. but at least one element of B that is not the part of A. $A \subset B$	$A = \{1,2,3,5\}$ $B = \{1,2,3,4,5\}$ $A \subset B = \{1,2, 3, 5\}$
Equal Set	If every element of A in B and every element of B in A. $A = B, A \subseteq B, B \subseteq A$	$A = \{2,4,6,6,10\}$ $B = \{2,4,6,6,10\}$ $A = B$
Null Set	A set which contains no element is called null set.	$A = \{ \}$
Universal Set	The set of all elements under consideration is called the Universal Set.	$U = \{1,2,3,4,\dots,100\}$
Finite Set	Finite sets are sets having a finite/countable number of members. Finite sets are also known as countable sets	$A = \{\text{Month of year}\}$
Infinite Set	An infinite set is a set whose elements can not be counted. An infinite set is one that has no last element.	$A = \{\text{even integers}\}$
Union	The set of all the elements which lie in set A and set B or both the elements in A and B altogether.	$A = \{1,3,5,7\}$ $B = \{2,4,6,8\}$ $A \cup B = \{1,2,3,4,5,6,7,8\}$
Intersection	Common element of both set A and B,	$A = \{1,3,5,7\}$ $B = \{2,3,4,5,6,8\}$ $A \cap B = \{3,5\}$

Partition of set	A set may divide up into some disjoint subsets. It is called partition of set.	$A = A \cup A_1 \cup A_2 \dots A_N$
Power Set	The power set of a set A is the set of all subsets of A, denoted by P(A)	$A = \{1, 2\}$ $P(A) = (\{\}, \{1\}, \{2\}, \{1, 2\})$
Ordered Pair	An ordered pair (a, b) consists of two elements “a” and “b” in which “a” is the first element and “b” is the second element	$A = \{1, 2, 3\}$ and a set $B = \{a, b, c\}$. The ordered set pair (A, B) would be represented as: $\{(1, a), (2, b), (3, c)\}$ In this example, the first element of each pair comes from set A, and the second element comes from set B.
Cartesian Product of two set	Let A and B be sets. The Cartesian product of A and B, denoted $A \times B$ (read “A cross B”) is the set of all ordered pairs (a, b), where a is in A and b is in B.	$A = \{1, 2\}$ and set $B = \{a, b, c\}$. $A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$ $B \times A = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}$

Membership Table

Definition	Representation	
A table displaying the membership of elements in sets	To indicate that an element is in a set, a 1 is used	to indicate that an element is not in a set, a 0 is used

Set Identities

Identity	Name
$A \cap B = B \cap A$ $A \cup B = B \cup A$	Commutative Laws
$(A \cap B) \cap C = A \cap (B \cap C)$ $(A \cup B) \cup C = A \cup (B \cup C)$	Associative Laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive Laws
$A \cap U = A$ $A \cup U = U$	Identity Law (Intersection and Union with Universal Set)
$(A')' = A$	Double Complement Laws
$A \cap A = A$ $A \cup A = A$	Idempotent Laws
$(A \cap B)' = A' \cup B'$ $(A \cup B)' = A' \cap B'$	De Morgan's Laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption Laws
$A - B = A \cap B'$	Set Difference Law